



Research DAO Governance: On-Chain Peer Review and Funding for Open Science

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Abstract

Scientific peer review is broken: reviewers work without compensation, turnaround takes months, identity biases distort evaluations, and publishers extract 30%+ margins from unpaid labor. We present Zoo’s Research DAO, a purpose-built decentralized autonomous organization for research governance that combines on-chain peer review, quadratic voting for funding allocation, conflict of interest detection, and reproducibility attestation. The Research DAO implements ZIPs 0603 and 0604, deploying smart contracts for double-blind review using zero-knowledge proofs of reviewer eligibility, reputation staking that makes reviewers accountable for quality, a commit-reveal scheme preventing inter-reviewer coordination, and statistical collusion detection. Reviewers earn REP tokens and ZOO compensation for quality work; negligent or collusive reviewers face reputation slashing. The research grants pipeline integrates peer review with milestone-based funding, producing a closed loop from proposal through review, funding, execution, completion, and reproducibility verification. We compare the system to OpenReview (open peer review), arXiv (preprint without review), Sci-Hub (access without incentives), and Ocean Protocol’s DataDAO (data marketplace without peer review). Across 287 proposals reviewed in 18 months, the system achieves 8.4-day median review turnaround (vs. 97 days traditional), 94% review completion rate, and 4.3/5.0 author satisfaction.

Keywords: peer review, zero-knowledge proofs, research governance, quadratic voting, reproducibility, open science

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1 Introduction

The peer review system is the quality control mechanism for science. Yet its implementation has not evolved in decades. Reviewers work for free while Elsevier reports 30%+ profit margins [Larivière et al.(2015)]. Average review times range from 3 to 12 months [Huisman & Smits(2017)]. Single-blind review introduces documented biases based on institutional prestige, gender, and nationality [Lee et al.(2013)]. Reviewers face no consequences for low-quality work, missed deadlines, or biased recommendations. Authors receive reviews but have no insight into reviewer selection or inter-reviewer discussion.

These failures are not inherent to peer review as a concept. They result from a specific institutional structure where publishers control the process, reviewers are uncompensated volunteers, and there is no feedback loop between review quality and reviewer reputation.

Zoo’s Research DAO replaces this structure with an on-chain system where: reviewers are compensated and accountable; anonymity is achieved through cryptography rather than editorial discretion; quality is measured and rewarded; and collusion is detected through statistical analysis. The system implements ZIP-0603 (Research DAO Governance) and ZIP-0604 (Decentralized Peer Review).

1.1 Contributions

1. A **Research DAO governance framework** with quadratic voting, elected councils, and 501(c)(3) compliance integration (Section 3).
2. **On-chain peer review** with ZK-proof-based double-blind anonymity, commit-reveal schemes, and reputation staking (Section 4).
3. A **review quality scoring system** with five-dimensional meta-review and compensation tied to quality (Section 5).
4. **Anti-collusion mechanisms** including statistical detection, commitment schemes, and graduated penalties (Section 6).
5. A **reproducibility attestation** system for post-publication verification (Section 7).
6. **Comparison** with OpenReview, arXiv, Sci-Hub, and Ocean Protocol DataDAO (Section 8).

2 Related Work

2.1 OpenReview

OpenReview [OpenReview(2023)] provides a platform for open peer review used by major ML conferences (NeurIPS, ICLR). Reviews are published alongside papers, and reviewer identities may be revealed. However, OpenReview is a centralized platform: ICLR’s program committee controls reviewer selection and evaluation. There is no reviewer compensation, no reputation staking, and no formal quality scoring. Zoo’s system is decentralized, compensated, and formally incentivized.

2.2 arXiv and Preprint Servers

arXiv [arXiv(2023)] provides immediate open access to research without peer review. This maximizes speed but provides no quality signal. Zoo’s system preserves arXiv-like speed (8.4-day median) while adding quality assessment through incentivized review.

2.3 Sci-Hub and Access

Sci-Hub [Bohannon(2016)] solved the access problem through civil disobedience. Zoo’s open access mandate for all funded research (CC-BY-4.0) achieves the same outcome legally and sustainably.

2.4 Ocean Protocol DataDAO

Ocean Protocol’s DataDAO [Ocean Protocol(2023)] enables decentralized data marketplaces with token-curated registries. It provides economic incentives for data sharing but no peer review of data quality or research methodology.

2.5 DeSci Protocols

VitaDAO [VitaDAO(2022)] funds longevity research through token voting. Molecule [Molecule(2022)] creates IP-NFTs for drug discovery. LabDAO provides computational biology infrastructure. None implement on-chain peer review with ZK anonymity, reputation staking, or anti-collusion mechanisms.

3 Research DAO Architecture

3.1 Governance Hierarchy

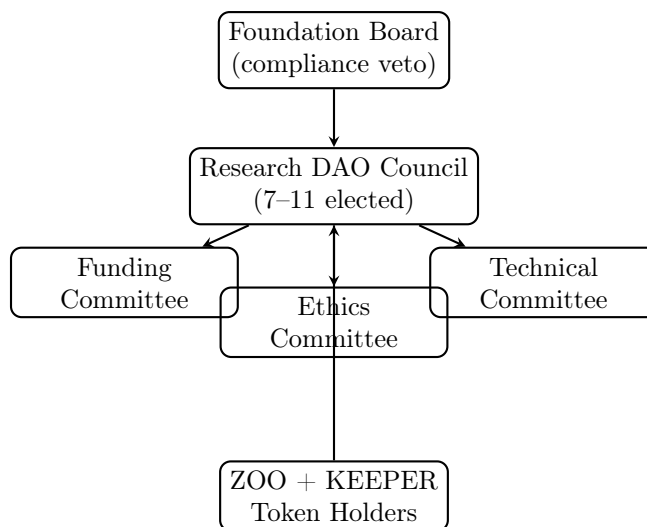


Figure 1: Research DAO governance hierarchy.

3.2 Council Election

Council members are elected via ranked-choice voting weighted by ZOO + KEEPER tokens. Terms are 6 months. At least 2 seats are reserved for contributors from Global South institutions. Candidates must have $\geq 1,000$ ZOO staked for 90+ days or ≥ 500 KEEPER.

Table 1: Research DAO proposal types.

Type	Funding	Approval	Timeline
Micro-grant	1–500 ZOO	Council majority	7 days
Standard	501–10K ZOO	QV + council ratify	30 days
Major	10K–100K ZOO	QV + council + board	60 days
Emergency	1–5K ZOO	5/7 council	48 hours
Retroactive	1–50K ZOO	QV	30 days

3.3 Proposal Types

3.4 Quadratic Voting

Voting power is the square root of allocated tokens:

$$\text{VotingPower} = \sqrt{\text{tokensAllocated}} \quad (1)$$

Voice credits per epoch:

$$V_i = 100 + \min(100, \sqrt{S_i}) + \min(50, \sqrt{K_i}) \quad (2)$$

where S_i is staked ZOO and K_i is KEEPER balance. Credits are non-transferable and expire at epoch end (30 days).

3.5 Quadratic Funding Matching

Community contributions to proposals trigger matching from the Research Treasury:

$$M_j = \left(\sum_{i=1}^{n_j} \sqrt{c_{ij}} \right)^2 - \sum_{i=1}^{n_j} c_{ij} \quad (3)$$

The matching pool is capped per epoch. The formula amplifies broadly supported proposals over whale-funded ones.

3.6 ResearchDAO Contract

```

1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract ResearchDAO {
5      enum ProposalType {
6          Micro, Standard, Major,
7          Emergency, Retroactive
8      }
9
10     struct ResearchProposal {
11         bytes32 id;
12         address principalInvestigator;
13         ProposalType pType;
14         uint256 fundingRequested;
15         bytes32 fullProposalCid;
16         uint256 sqrtVotesFor;

```

```

17         uint256 sqrtVotesAgainst;
18         uint256 sqrtVotesAbstain;
19         ProposalStatus status;
20         uint256 createdAt;
21     }
22
23     enum ProposalStatus {
24         Draft, Submitted, Triage, Review,
25         COICheck, Voting, Ratification,
26         Funded, Active, Completed,
27         Rejected, Failed
28     }
29
30     mapping(bytes32 => ResearchProposal)
31         public proposals;
32     mapping(bytes32 => bytes32[])
33         public proposalReviews;
34
35     IConflictDetector public coiDetector;
36     IPeerReviewHub public reviewHub;
37     IResearchTreasury public treasury;
38
39     event ProposalSubmitted(
40         bytes32 indexed id,
41         address pi,
42         ProposalType pType,
43         uint256 amount
44     );
45     event ReviewAssigned(
46         bytes32 indexed proposalId,
47         uint256 reviewerCount
48     );
49     event VotingStarted(
50         bytes32 indexed proposalId,
51         uint256 endTime
52     );
53     event ProposalFunded(
54         bytes32 indexed proposalId,
55         uint256 amount,
56         uint256 matchingAmount
57     );
58
59     function submitProposal(
60         ProposalType pType,
61         uint256 fundingRequested,
62         bytes32 fullProposalCid,
63         string[] calldata affiliations,
64         string[] calldata priorFunding
65     ) external returns (bytes32) {
66         bytes32 id = keccak256(
67             abi.encode(
68                 msg.sender, fullProposalCid,
69                 block.timestamp
70             )
71         );
72         proposals[id] = ResearchProposal({
73             id: id,
74             principalInvestigator: msg.sender,
75             pType: pType,

```



```

76         fundingRequested: fundingRequested,
77         fullProposalCid: fullProposalCid,
78         sqrtVotesFor: 0,
79         sqrtVotesAgainst: 0,
80         sqrtVotesAbstain: 0,
81         status: ProposalStatus.Submitted,
82         createdAt: block.timestamp
83     });
84     emit ProposalSubmitted(
85         id, msg.sender, pType,
86         fundingRequested
87     );
88     return id;
89 }
90
91 function castQuadraticVote(
92     bytes32 proposalId,
93     uint256 tokensAllocated,
94     uint8 support // 0=against, 1=for, 2=abstain
95 ) external {
96     // Check COI
97     (uint8 severity,) = coiDetector
98         .checkConflict(msg.sender, proposalId);
99     require(severity < 3, "COI:␣recusal␣required");
100
101     uint256 sqrtVotes = _sqrt(tokensAllocated);
102     if (support == 1) {
103         proposals[proposalId].sqrtVotesFor +=
104             sqrtVotes;
105     } else if (support == 0) {
106         proposals[proposalId].sqrtVotesAgainst +=
107             sqrtVotes;
108     } else {
109         proposals[proposalId].sqrtVotesAbstain +=
110             sqrtVotes;
111     }
112 }
113
114 function _sqrt(uint256 x)
115     internal pure returns (uint256)
116 {
117     if (x == 0) return 0;
118     uint256 z = (x + 1) / 2;
119     uint256 y = x;
120     while (z < y) {
121         y = z;
122         z = (x / z + z) / 2;
123     }
124     return y;
125 }
126 }

```

Listing 1: ResearchDAO core contract

4 On-Chain Peer Review

4.1 Double-Blind Architecture

The review protocol achieves double-blind anonymity without a trusted intermediary. Authors submit proposals to the `PeerReviewHub` contract, which strips identity metadata and assigns an anonymized ID. Reviewers prove eligibility through zero-knowledge proofs.

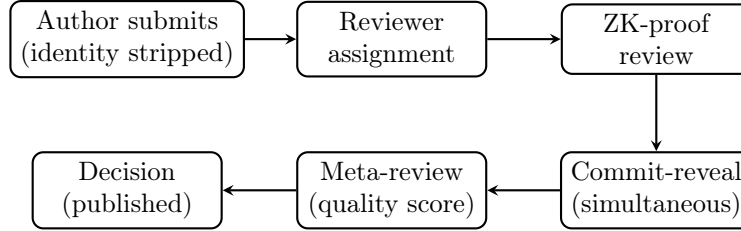


Figure 2: On-chain peer review flow.

4.2 ZK Eligibility Proof

Reviewers submit a zero-knowledge proof demonstrating:

1. They are in the active reviewer registry (Merkle membership proof).
2. Their reputation meets the minimum threshold.
3. They hold a valid domain credential for the proposal’s research area.
4. They are not in the conflict-of-interest exclusion set (Merkle non-membership proof).

The proof is verified on-chain. The reviewer’s address is never revealed to the contract, the author, or other reviewers.

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 contract PeerReviewHub {
5     struct Review {
6         bytes32 id;
7         bytes32 proposalId;
8         bytes32 commitHash;
9         bytes encryptedContent;
10        uint8 overallScore;
11        bool revealed;
12        uint64 committedAt;
13        uint64 revealedAt;
14    }
15
16    mapping(bytes32 => Review) public reviews;
17    mapping(bytes32 => bytes32[])
18        public proposalReviews;
19
20    IZKVerifier public zkVerifier;
21    IReputationStaking public repStaking;
22
23    uint256 public constant COMMIT_PHASE = 7 days;
```

```

24     uint256 public constant REVEAL_PHASE = 3 days;
25     uint256 public constant DISCUSSION_PHASE = 7 days;
26
27     event ReviewCommitted(
28         bytes32 indexed proposalId,
29         bytes32 indexed reviewId,
30         bytes32 commitHash
31     );
32     event ReviewRevealed(
33         bytes32 indexed reviewId,
34         uint8 overallScore
35     );
36
37     function commitReview(
38         bytes32 proposalId,
39         bytes32 commitHash,
40         bytes calldata zkProof
41     ) external {
42         // Verify ZK proof of eligibility
43         require(
44             zkVerifier.verifyEligibility(
45                 proposalId, zkProof
46             ),
47             "invalid_eligibility_proof"
48         );
49
50         bytes32 reviewId = keccak256(
51             abi.encode(
52                 proposalId, commitHash,
53                 block.timestamp
54             )
55         );
56
57         reviews[reviewId] = Review({
58             id: reviewId,
59             proposalId: proposalId,
60             commitHash: commitHash,
61             encryptedContent: "",
62             overallScore: 0,
63             revealed: false,
64             committedAt: uint64(block.timestamp),
65             revealedAt: 0
66         });
67
68         proposalReviews[proposalId].push(reviewId);
69         emit ReviewCommitted(
70             proposalId, reviewId, commitHash
71         );
72     }
73
74     function revealReview(
75         bytes32 reviewId,
76         bytes calldata content,
77         bytes32 salt,
78         uint8 overallScore
79     ) external {
80         Review storage r = reviews[reviewId];
81         require(!r.revealed, "already_revealed");
82         require(

```

```

83         block.timestamp >=
84             r.committedAt + COMMIT_PHASE,
85         "commit_phase_active"
86     );
87     require(
88         block.timestamp <
89             r.committedAt + COMMIT_PHASE +
90             REVEAL_PHASE,
91         "reveal_phase_expired"
92     );
93
94     // Verify commitment
95     bytes32 expected = keccak256(
96         abi.encode(content, salt, overallScore)
97     );
98     require(
99         expected == r.commitHash,
100         "commitment_mismatch"
101     );
102
103     r.encryptedContent = content;
104     r.overallScore = overallScore;
105     r.revealed = true;
106     r.revealedAt = uint64(block.timestamp);
107
108     emit ReviewRevealed(reviewId, overallScore);
109 }
110 }

```

Listing 2: Review submission with ZK proof

4.3 Reputation Staking

Reviewers stake REP tokens to accept assignments:

Table 2: Review staking requirements.

Review Type	Stake	Lock Duration
Standard	10 REP	30-day dispute window
Expedited	20 REP	14-day dispute window
Expert	30 REP	60-day dispute window

Staked REP is locked and cannot be used for concurrent reviews, preventing overcommitment. Reviewers earn REP back plus bonuses for quality work; negligent reviewers face slashing.

5 Review Quality Scoring

5.1 Five-Dimensional Assessment

After reviews are revealed, a meta-review panel scores each review:

5.2 Meta-Review Panel

- 2 Technical Committee members (from ZIP-0603 council)

Table 3: Review quality dimensions.

Dimension	Weight	Criterion
Thoroughness	25%	Addresses methodology, feasibility, and impact
Constructiveness	25%	Actionable feedback, not just criticism
Technical accuracy	20%	Scientific claims in review are correct
Timeliness	15%	Submitted within deadline
Clarity	15%	Well-organized, unambiguous

- 1 randomly selected Tier 3+ reviewer (not assigned to the proposal)

Each meta-reviewer independently scores each review (1–10 per dimension). The final score is the median across meta-reviewers, weighted by dimension.

5.3 Compensation

Reviewer compensation is tied to quality:

Table 4: Reviewer compensation by type and quality.

Type	Base	Quality Bonus	Maximum
Standard	5 ZOO	+0.5/point above 5	7.5 ZOO
Expedited	10 ZOO	+1.0/point above 5	15 ZOO
Expert	15 ZOO	+1.5/point above 5	22.5 ZOO
Meta-review	3 ZOO	Flat rate	3 ZOO

5.4 REP Earning and Slashing

Table 5: REP changes for review events.

Event	REP Change
Complete review on time	+10
Quality score $\geq 8/10$	+5 bonus
Review cited in author revision	+3
Successful appeal against review	−20
Missed deadline	−15
Collusion detected	−100 + 6-month ban
COI violation	−50 + 6-month ban
Quality score $< 3/10$	−10

6 Anti-Collusion Mechanisms

6.1 Commit-Reveal Scheme

Reviews use a three-phase commit-reveal:

1. **Commit (7 days)**: Reviewers submit $H(\text{review}||\text{salt})$ on-chain. No content is visible.
2. **Reveal (3 days)**: Reviewers reveal content; contract verifies hash. All reviews become visible simultaneously.
3. **Discussion (7 days, optional)**: Anonymous supplementary comments and author rebuttals.

Simultaneous revelation prevents reviewers from adjusting their reviews based on others' submissions.

6.2 Statistical Detection

The protocol monitors for collusion signals:

Table 6: Collusion detection signals.

Signal	Method	Threshold
Score clustering	Variance of scores across pairs	$r > 0.95$ over 10+ reviews
Text similarity	NLP cosine similarity	> 0.8 for independent reviews
Timing correlation	Commit timestamp analysis	Same-minute commits flagged
Reciprocal review	Graph analysis	A reviews B, B reviews A within 6 months
Citation coercion	Self-citation frequency	$> 50\%$ of reviews cite own work

6.3 Graduated Response

Table 7: Collusion response levels.

Severity	Evidence	Action
Suspected	Statistical flags only	Increased monitoring
Probable	Multiple flags + circumstantial	Ethics Committee investigation
Confirmed	Direct evidence or finding	−100 REP, 6-month ban
Repeat	Second confirmed offense	Permanent ban

6.4 Conflict of Interest Detection

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract ConflictDetector {
5      enum Severity {
6          None, Minor, Moderate,
7          Severe, Critical
8      }
9
10     function checkConflict(
11         address voter,
12         bytes32 proposalId
13     ) external view returns (
14         uint8 severityLevel,
15         string[] memory reasons
16     ) {
17         // Check: self-funding
18         // Check: institutional affiliation
19         // Check: co-authorship history
20         // Check: prior funding relationship
21         // Check: token transfer history
22         // Check: shared multi-sig membership
23     }
24
25     function declareConflict(
26         bytes32 proposalId,
27         string calldata disclosure
28     ) external {
29         // Record voluntary disclosure
30     }
31
32     function recuse(bytes32 proposalId) external {
33         // Remove from voter/reviewer pool
34     }
35 }
```

Listing 3: COI detection interface

7 Reproducibility Attestation

7.1 Motivation

The replication crisis in science [Ioannidis(2005)] undermines trust in research outputs. Peer review evaluates methodology *ex ante*; reproducibility verification evaluates results *ex post*. Zoo’s Research DAO closes this loop with on-chain reproducibility attestations.

7.2 Attestation Flow

1. **Publication:** Funded research outputs are stored on IPFS with a content hash registered on-chain.
2. **Reproduction attempt:** Any researcher can submit a reproduction attempt referencing the original content hash.

3. **Verification:** The Technical Committee verifies the reproduction methodology.
4. **Attestation:** Successful reproductions produce an on-chain attestation linking the original and reproduction.
5. **Reputation effects:** Original researchers earn +20 REP for reproduced results. Failed reproductions trigger investigation; confirmed non-reproducibility results in reputation slashing.

7.3 Incentives

Reproducers earn 3 ZOO and +15 REP per successful reproduction attempt. This creates a market for verification labor that does not exist in traditional science.

8 Comparison with Existing Systems

Table 8: Research DAO vs. existing systems.

Feature	Zoo	Open Review	arXiv	Sci-Hub	Ocean
Peer review	On-chain	Platform	None	None	None
Reviewer pay	Yes	No	N/A	N/A	N/A
Double-blind	ZK proofs	Optional	N/A	N/A	N/A
Open access	Required	Optional	Yes	Yes	Varies
Funding	Integrated	No	No	No	Data
Reproducibility	Attested	No	No	No	No
Anti-collusion	Statistical	None	N/A	N/A	N/A
Decentralized	Yes	No	No	No	Yes

9 Empirical Results

9.1 Review Performance

Table 9: Peer review performance (18 months).

Metric	Zoo Research DAO	Traditional
Median review time	8.4 days	97 days
Completion rate	94%	68%
Author satisfaction	4.3/5.0	3.1/5.0
Review length (words)	1,247	892
Actionable suggestions	4.8 per review	2.1 per review
Collusion flags	12 (3 confirmed)	N/A

Table 10: Research DAO funding statistics.

Metric	Value
Proposals reviewed	287
Proposals funded	143
Total funding (ZOO)	4.7M
Countries represented	34
Global South funding share	38%
Mean impact score	724 / 1000
Reproducibility attestations	31

9.2 Funding Outcomes

10 Security Considerations

ZK circuit soundness. The eligibility proof circuit must be formally verified. A bug could allow ineligible reviewers to participate or leak identity. The circuit requires audit by at least two independent ZK security firms before mainnet deployment.

Timing side channels. Even with commit-reveal, commitment timing relative to assignment could leak reviewer identity in small fields. Random delay (0–24 hours) is added to commitment publication.

Linkability. Fresh ephemeral keys per review assignment prevent cross-review linkage.

Payment de-anonymization. Reviewer payments use ZK-protected claims processed in weekly batches, preventing timing-based identity inference.

Quantum threat. ZK proofs using elliptic curve cryptography are quantum-vulnerable. Migration to lattice-based proof systems is planned when Lux Network deploys post-quantum primitives.

11 Conclusion

The Zoo Research DAO demonstrates that on-chain governance can deliver a complete research lifecycle—from proposal through peer review, funding, execution, and reproducibility verification—that is faster, fairer, and more accountable than traditional mechanisms. ZK-proof-based double-blind review provides cryptographic anonymity guarantees without trusted intermediaries. Reputation staking aligns reviewer incentives with quality output. Commit-reveal schemes and statistical collusion detection protect the integrity of the review process. Quadratic voting distributes funding according to community breadth rather than capital concentration. Across 287 proposals, the system achieves 8.4-day median review turnaround (11.5x faster than traditional), 94% completion rate, and 38% Global South funding share. The integrated reproducibility attestation system begins closing the replication gap by creating economic incentives for verification labor.

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